

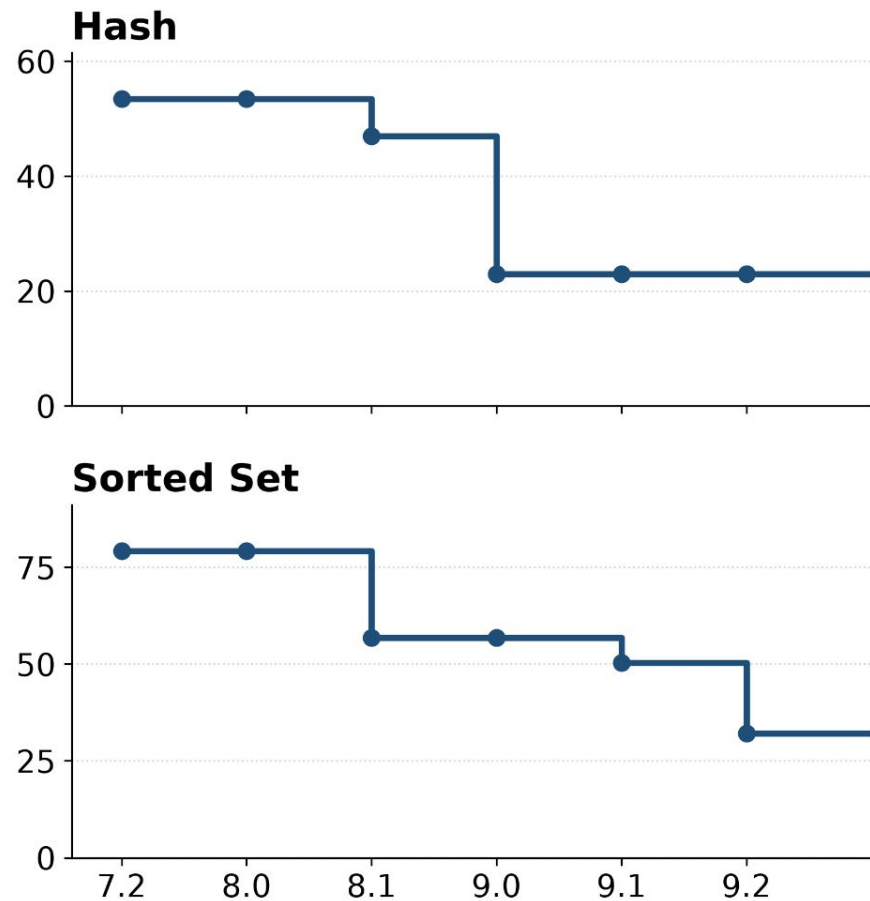
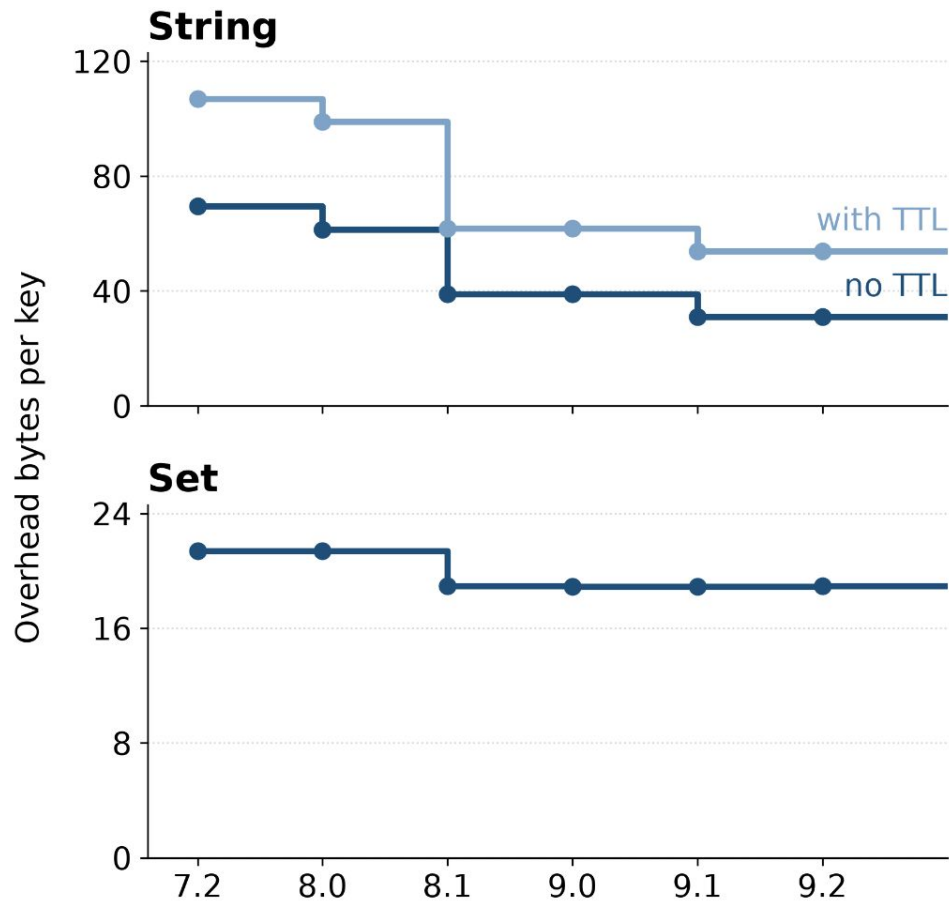
No Client Changes Required

Making Existing Workloads Faster with Less RAM

If I were to update Valkey 7.2 to 9.2...

- My dataset has 16B keys and 16B values
 - Memory usage drops 40%
 - 59% less overhead
-
- My dataset is large sorted sets with 20B elements
 - Memory usage drops 47%
 - 63% less overhead

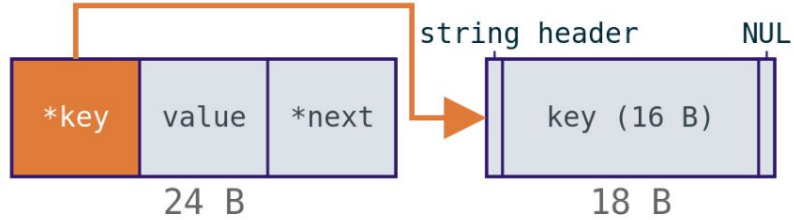
*Disclaimer: results vary due to allocator rounding



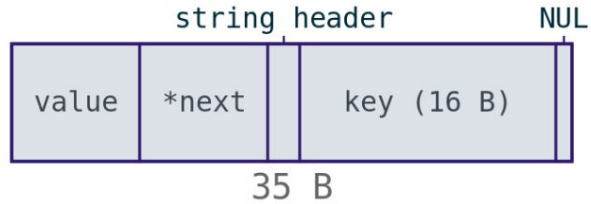
Embedding

Dictionary Entry

Before

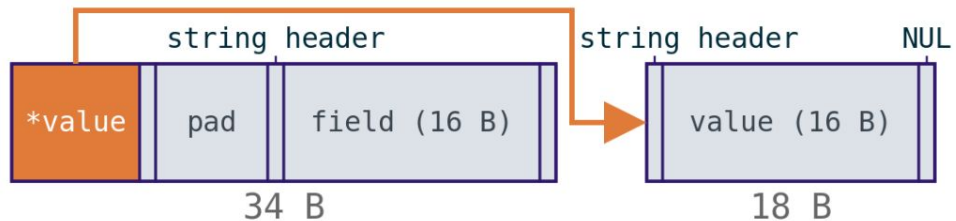


After

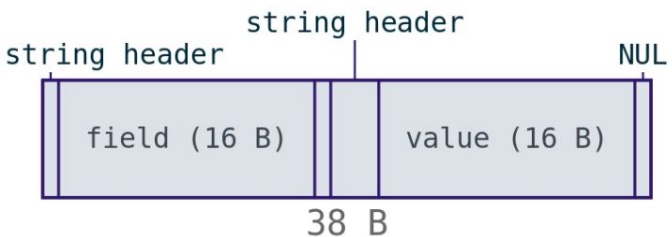


Hash entry

Before

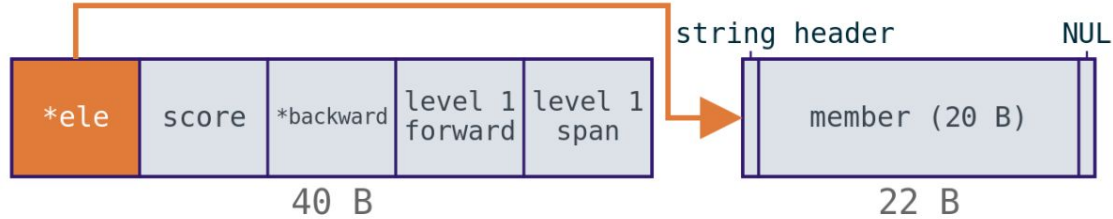


After

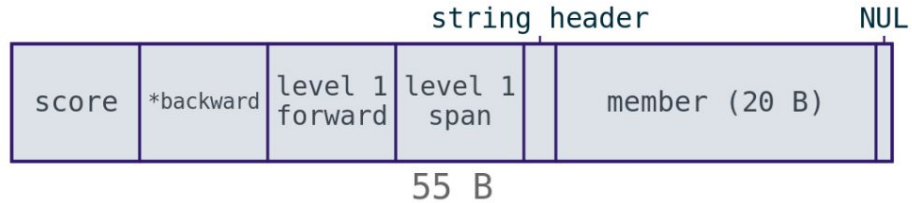


Sorted Set - Skiplist Node

Before

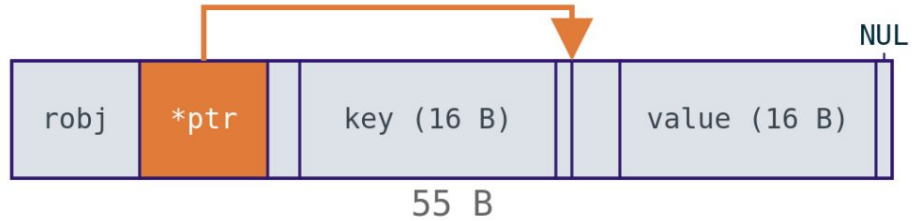


After

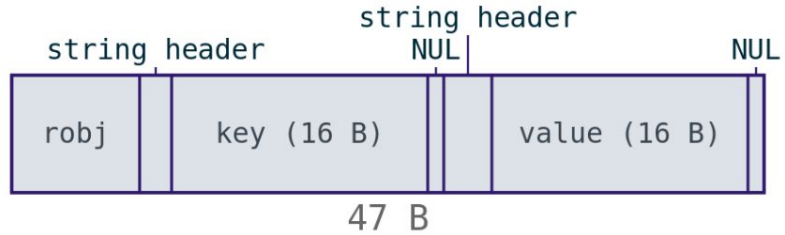


String object

Before



After

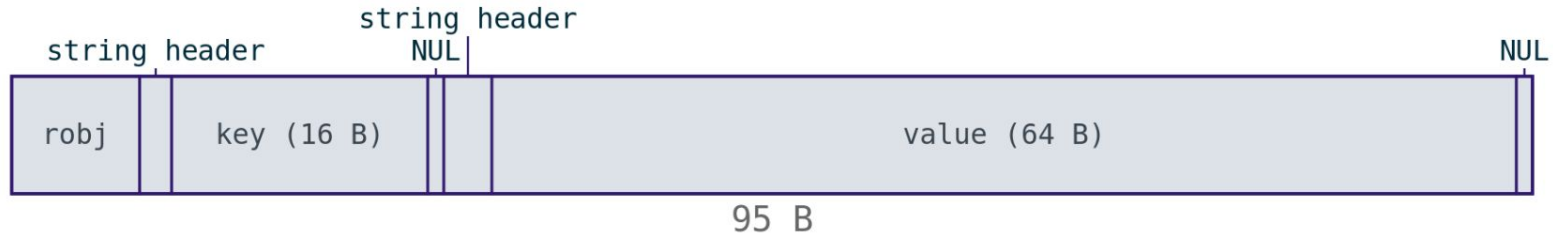


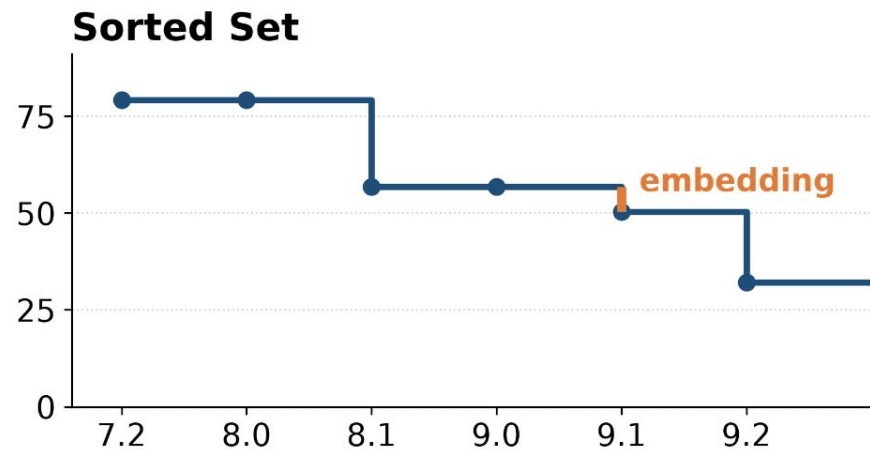
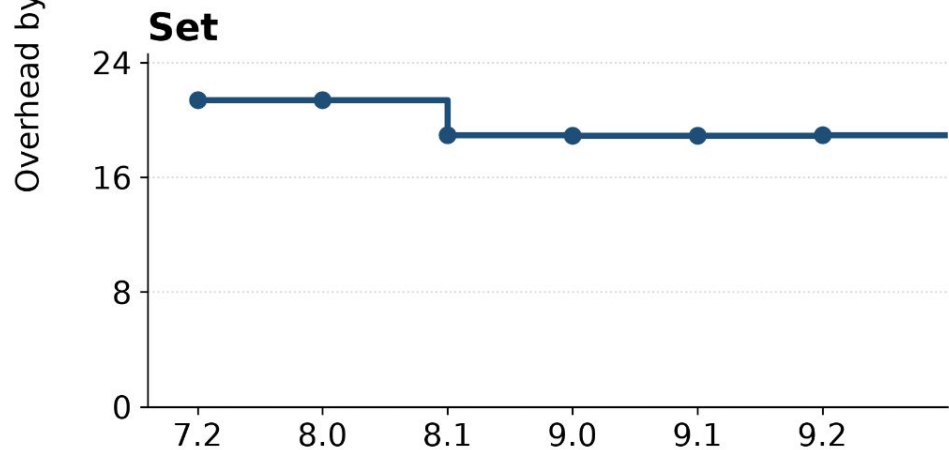
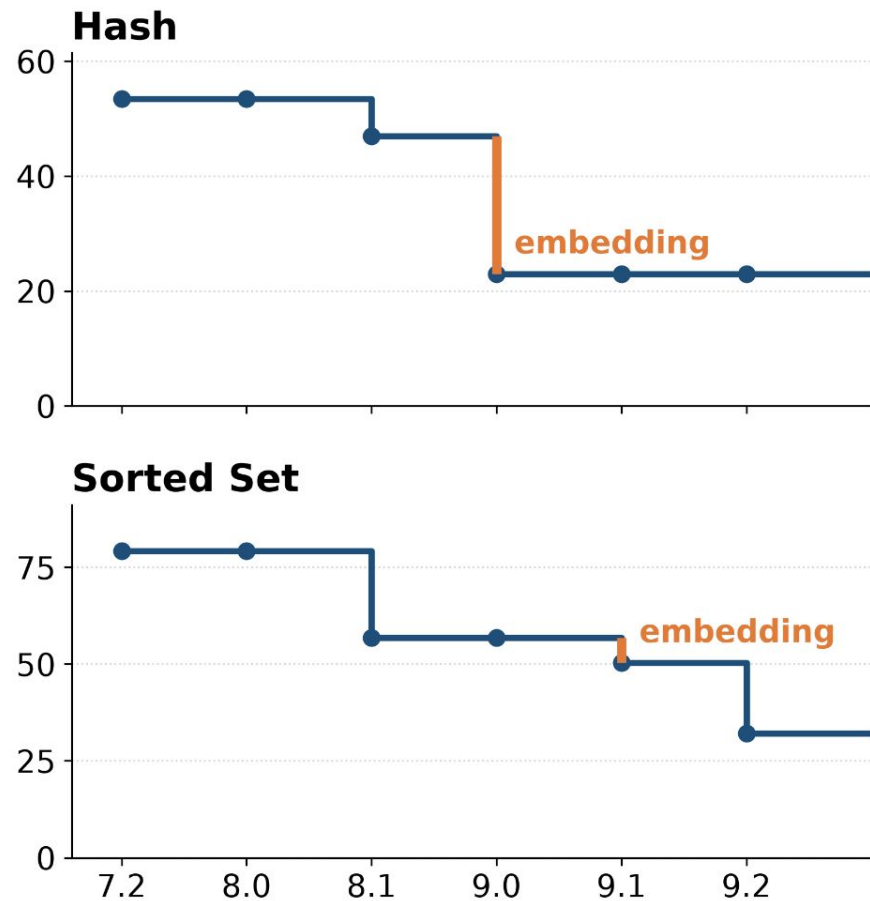
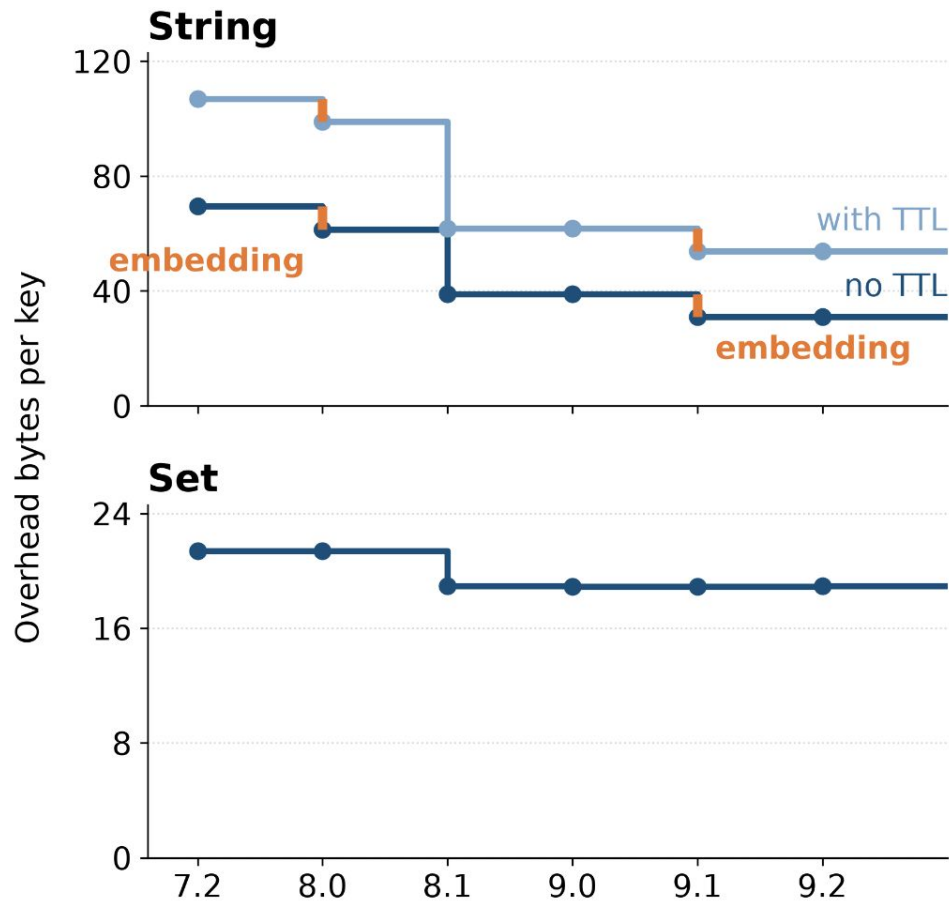
Raised string embedding threshold

Before



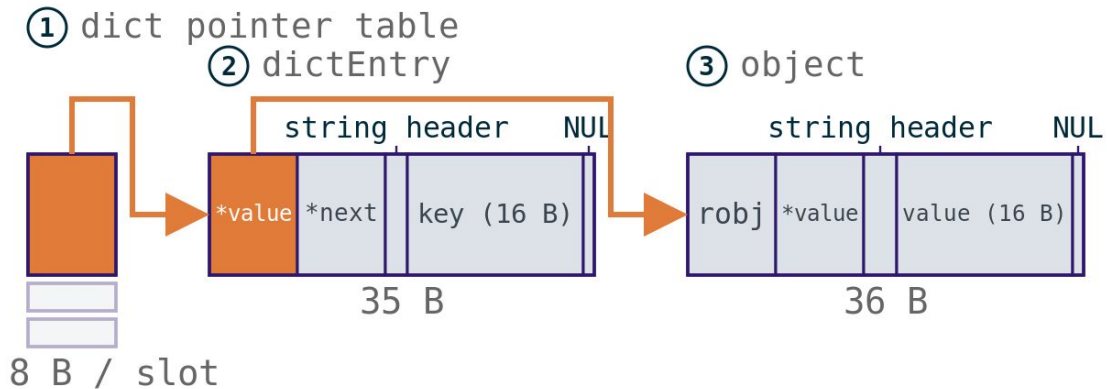
After



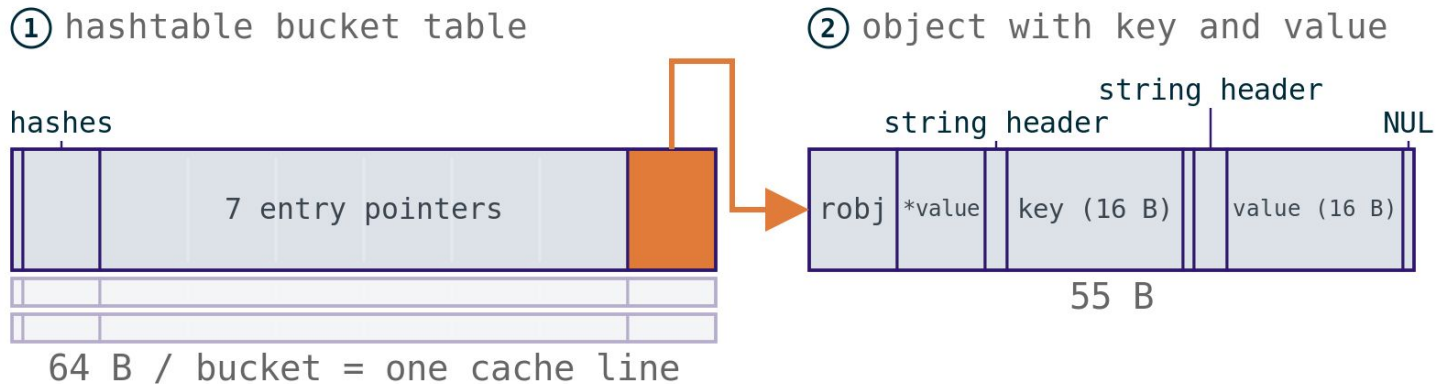


Dict ➡ Hashtable

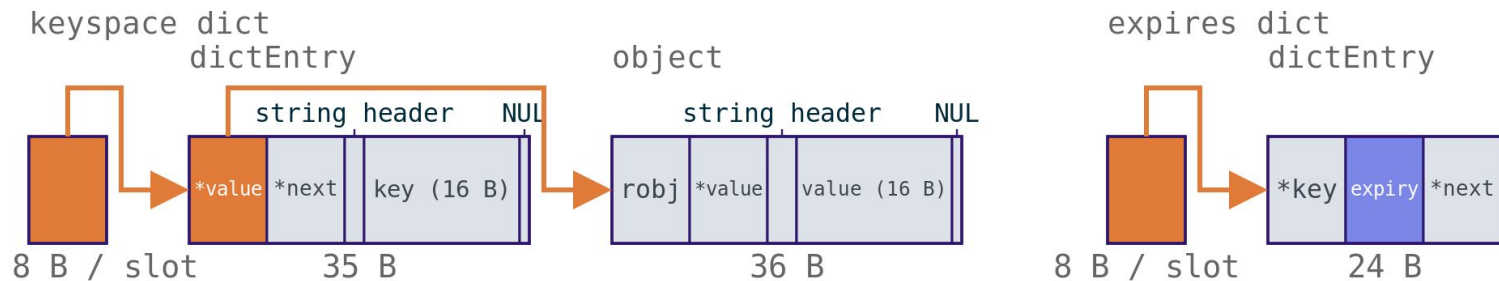
Before



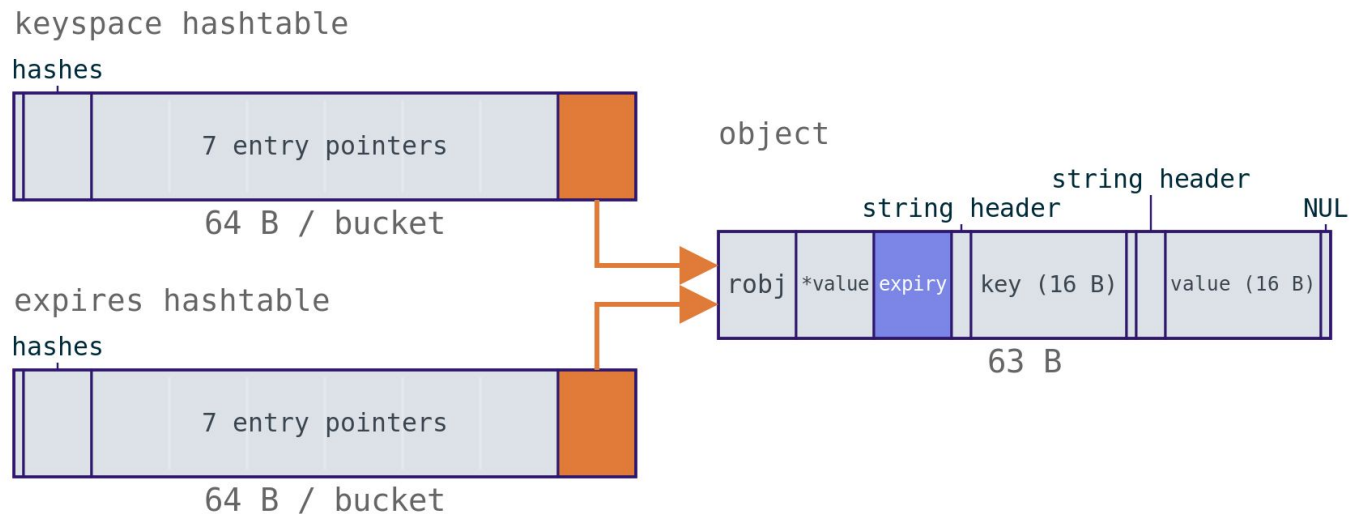
After

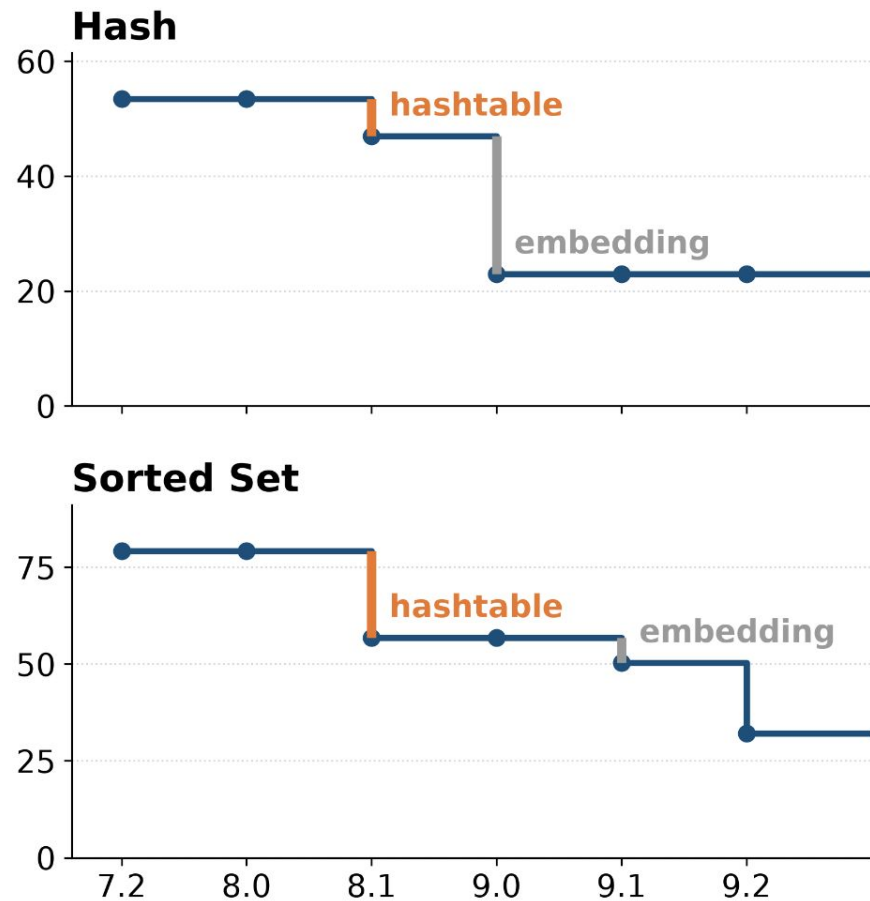
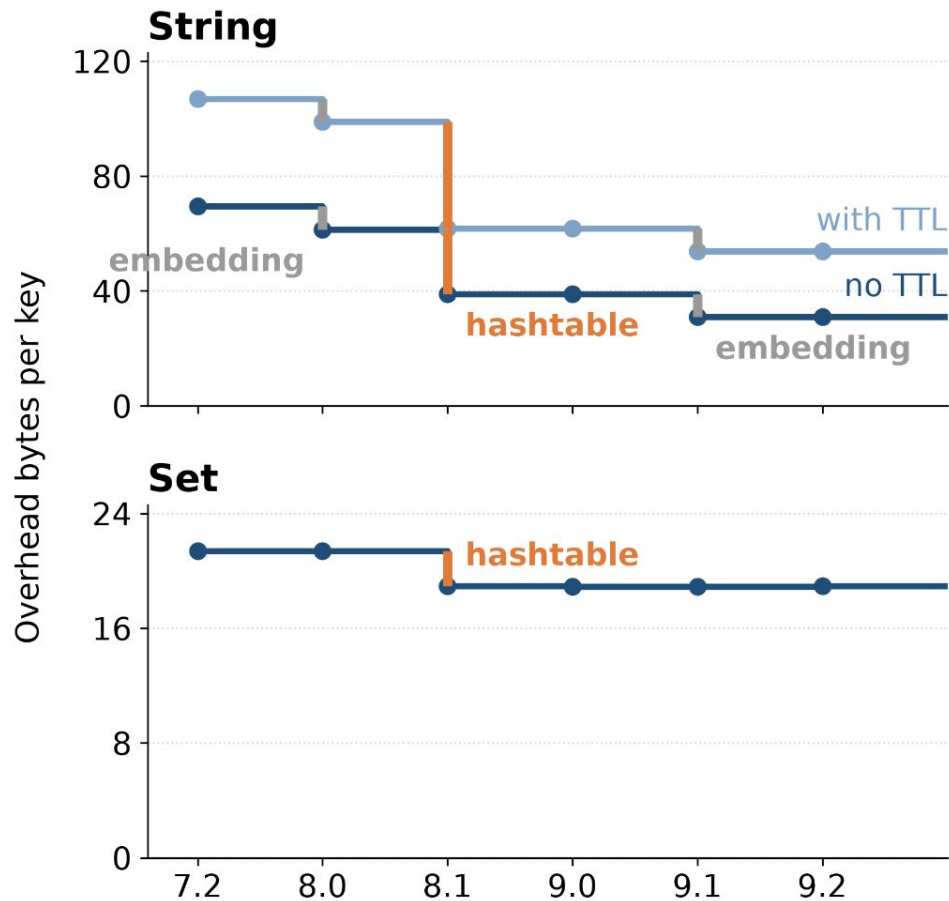


Before

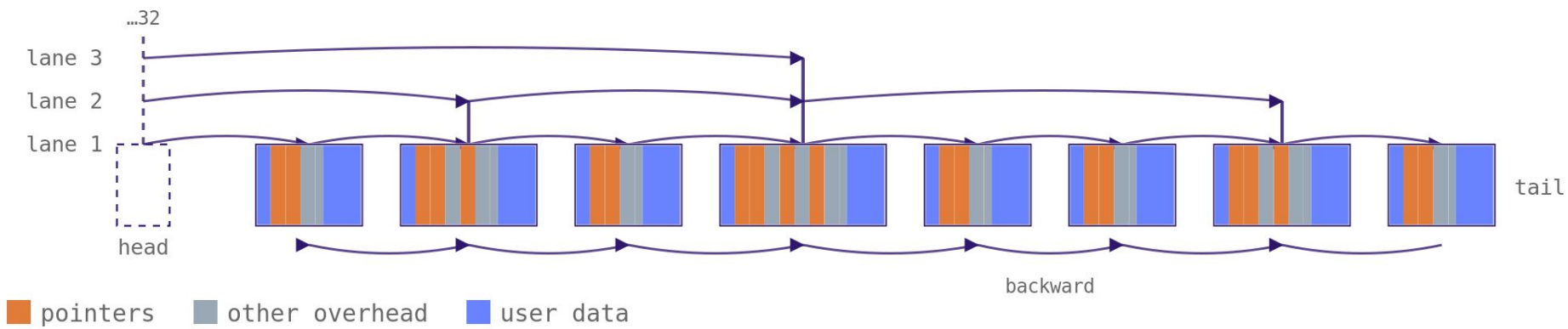


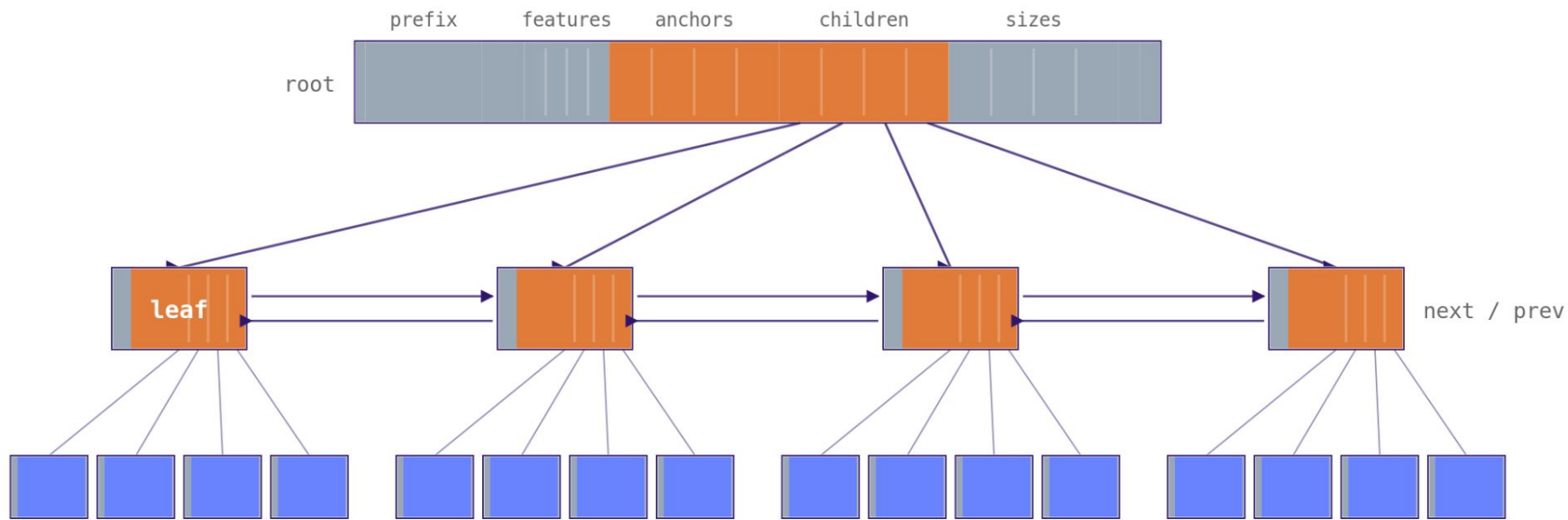
After





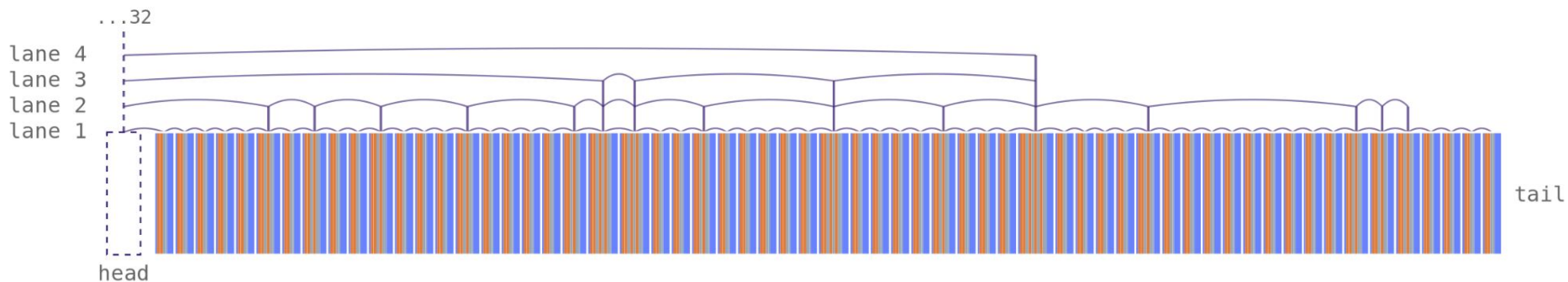
Skiplist ➡ B+ Tree



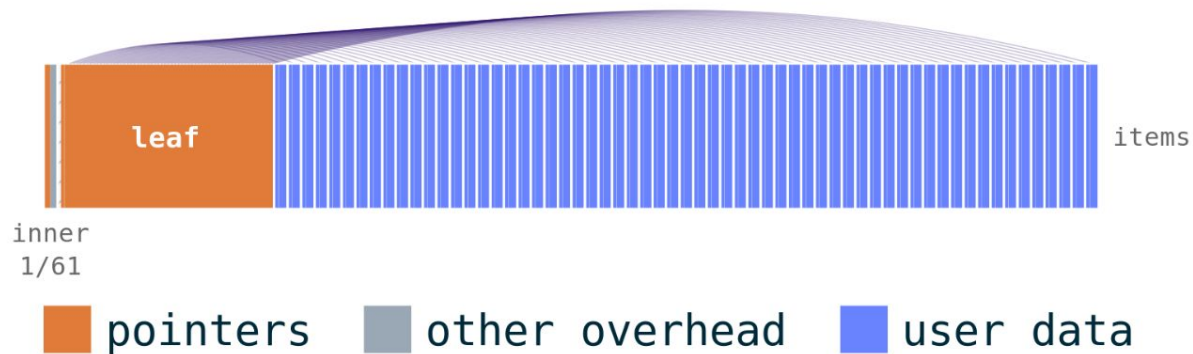


pointers other overhead user data

skiplist · 62.2 B/member

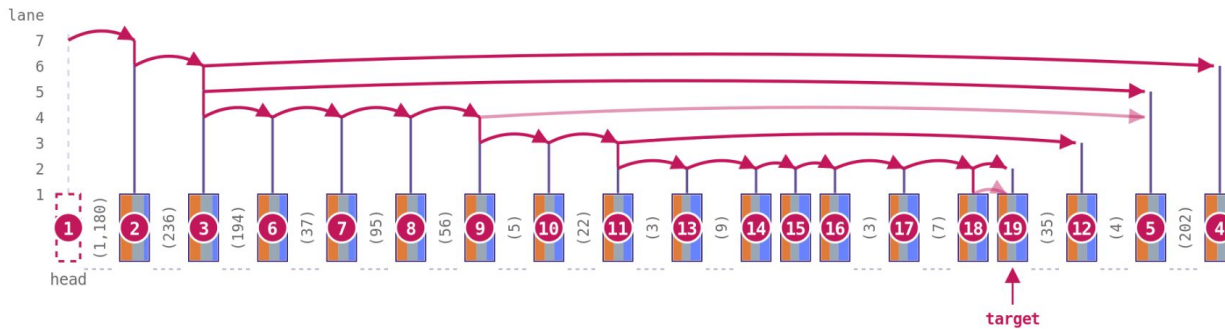


fbtree · 40.9 B/member



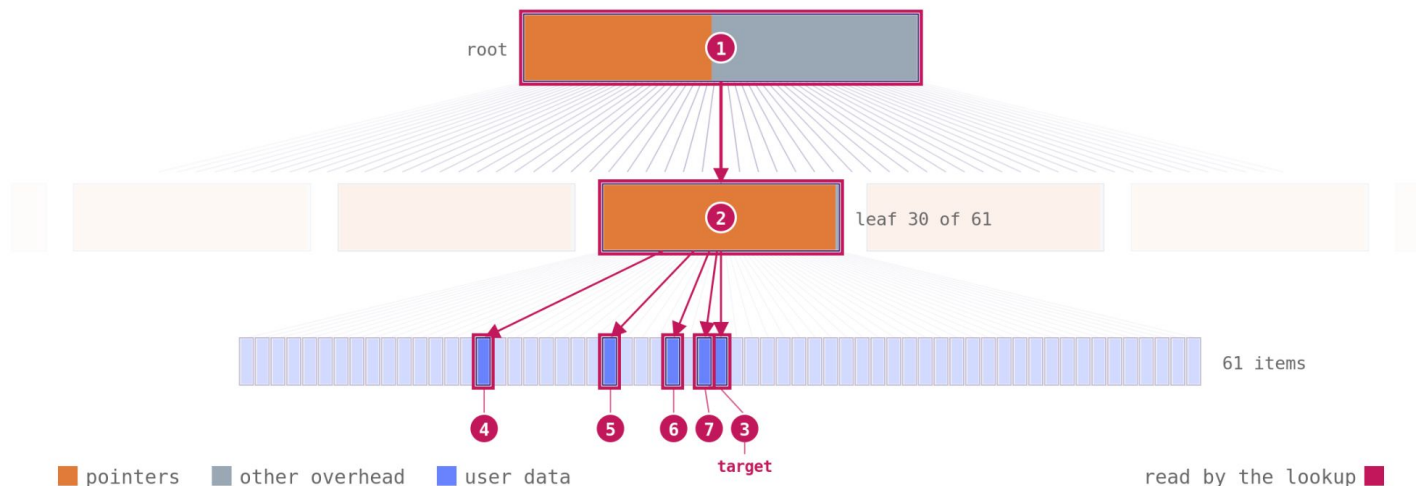
skiplist

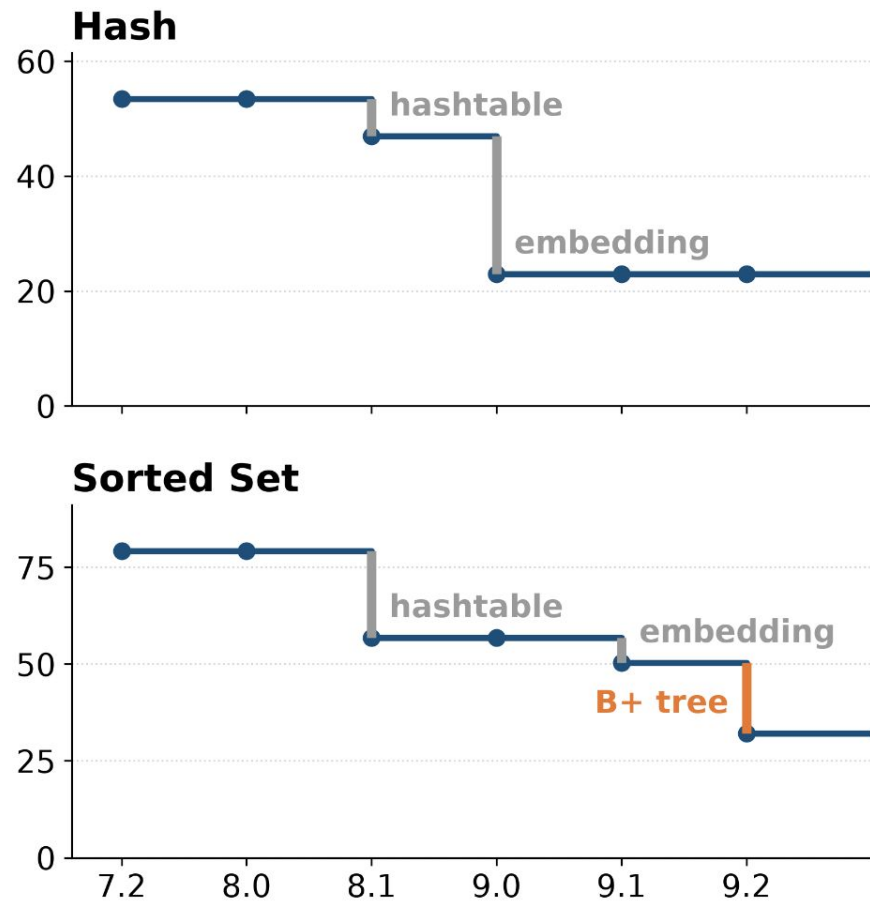
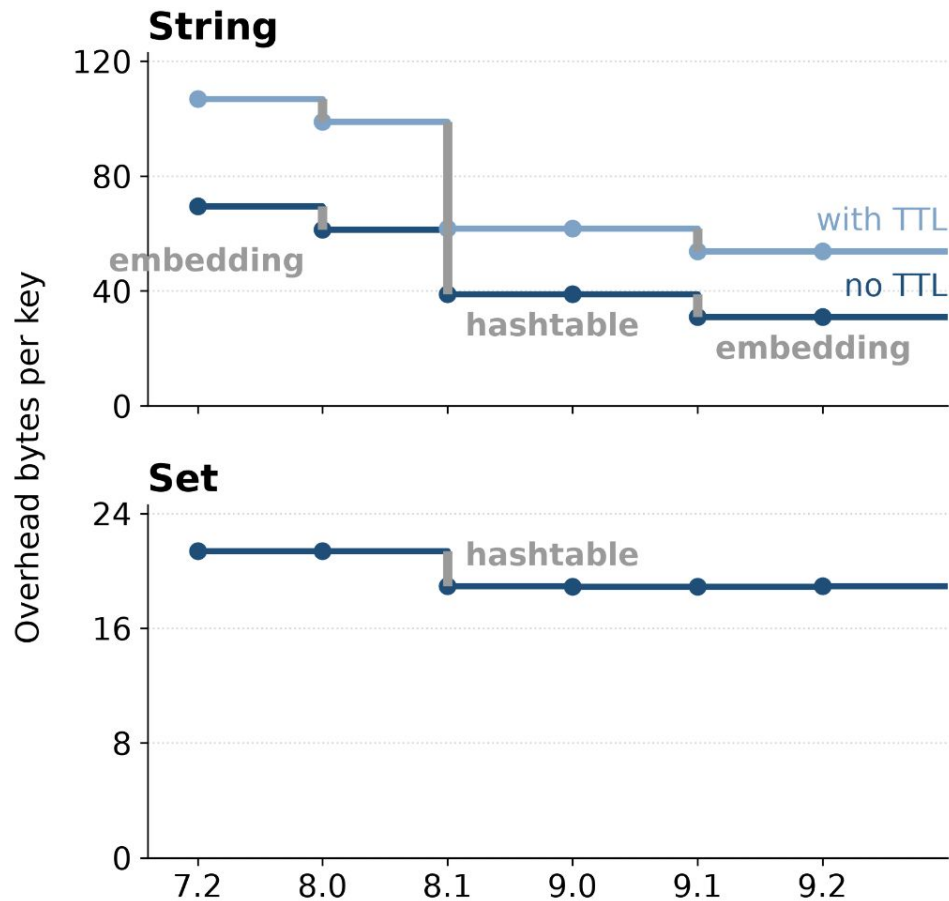
19 reads



fbtree

7 reads





The Future

RELEASE	PR	CHANGE	AUTHORS	REVIEWERS
8.0	541	Key embedded in dict entry	 Harkrishn Patro  Madelyn Olson	 Viktor Söderqvist  Ping Xie  Zvi Schneider
8.1	1186	New hashtable for the keyspace	 Viktor Söderqvist  Rain Valentine  Uri Yagelnik	 Madelyn Olson  Jim Brunner
8.1	1176, 1427, 1502	Hashtable for sets, sorted sets, hashes	 Rain Valentine	 Viktor Söderqvist  Ran Shidlansik
9.0	1579	Hash value embedded in entry	 Viktor Söderqvist	 Ran Shidlansik
9.1	2508	Member embedded in skiplist node	 chzhoo  Viktor Söderqvist	 Ran Shidlansik
9.1	2516	Value pointer dropped from string object	 Rain Valentine	 Viktor Söderqvist  Jim Brunner  Ran Shidlansik  Madelyn Olson
9.2	3397	Embedding limit 64 → 128 B	 Nikhil Manglore	 Viktor Söderqvist
9.2	3579, 3840, 4206, 4359	B+ tree for sorted sets	 Rain Valentine  Abhishek Kumar	 Eran Ifrah  Jim Brunner  Ran Shidlansik